Eyub Celebioglu

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Research Interests

Machine learning, reinforcement learning, computer vision and robotics. I plan to pursue a PhD.

Education

Université de Lorraine 2026 – Present
MSc in Artificial Intelligence, Vision & Robotics (second year) *Nancy, France*

- Coursework: deep learning, image processing, planning/learning/control for robotics, geometric models for vision, collective robotics.

Université de Lorraine 2025 – 2026
MSc in Computer Science, Artificial Intelligence track (first year) *Nancy, France*

- Coursework: research initiation, computer vision, autonomous mobile robotics, data parallelism, neuromorphic computing.

ISIMA – Université Clermont Auvergne 2022 – 2025
BSc in Computer Science *Clermont-Ferrand, France*

Université Clermont Auvergne 2021 – 2022
First year of BSc in Physics, then transferred to Computer Science *Clermont-Ferrand, France*

Research Experience

Autonomous Navigation in Uncertain Environments 2025 – 2026 (6 months)
First-year Master's research project, supervised by Olivier Buffet (Inria / LORIA) · team of 3 *Nancy, France*

- Built a 2D robot simulator (kinematics, 5-ray LIDAR) and implemented the RRT and RRT* path planners.
- Proposed **RRTSC**, a shortcut post-processing of RRT: ≈70% fewer nodes in the final path (14 vs. 55) and ≈6% lower cost than RRT, at a fraction of RRT* runtime (100 runs per setting, 2 maps).
- Trained a DQN agent in PyTorch (MLP 13-128-128-5, replay buffer, target network) to follow RRT paths while avoiding obstacles; 3000 episodes over 5 maps, 73–91% success depending on the map (90–100% in training).
- Diagnosed and fixed reward hacking (agent spinning in place) and miscalibrated penalties; oversampled rare collision transitions ×10 in the replay buffer to counter class imbalance.

Selected Projects

Neural Networks from Scratch | *C++, OpenMP* code

- Multi-layer perceptron and backpropagation without any ML library; matrix operations parallelised with OpenMP.

Deep Q-Learning for Snake | *Python, reinforcement learning* code

- Trained a DQN agent to play Snake from the game state; precursor to the navigation work.

Anime Face Generation with a DCGAN | *Python, GAN, Kaggle GPU* code

- Trained a deep convolutional generator and discriminator on GPU to generate anime character faces.

Local LLM Voice Assistant | *Python, Ollama, Docker* code

- Served open-source LLMs (Llama) locally in Docker containers, with a voice interface and speech synthesis.

3D Physics and Rendering Engine | *C++, OpenGL* code

- Modular engine with a custom .obj parser and real-time collisions; extended into a voxel engine with procedural terrain.

Technical Skills

Programming: Python, C/C++, Java, C#, SQL, R, Prolog
Machine learning: PyTorch, NumPy, Matplotlib; reinforcement learning (DQN), GANs
Robotics & systems: ROS, motion planning (RRT, RRT*), OpenMP, OpenGL, Docker, Linux, Git, CPLEX, SageMath
Languages: French (native), English (B2)